

Review

Recycled Materials in Construction: Trends, Status, and Future of Research

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Abstract: The utilization of recycled materials has emerged as a pivotal strategy for mitigating resource depletion and reducing carbon emissions in the construction industry. However, existing reviews predominantly focus on specific technical aspects, often overlooking the interdisciplinary complexities associated with recycled materials as a systems engineering challenge. This study systematically reviews 1533 documents from the Web of Science Core Collection, integrating quantitative and qualitative analytical approaches to assess the current state and future trajectory of the field, thereby addressing existing research gaps. The findings highlight the substantial evolution of recycled building materials from waste recovery to a multifaceted domain encompassing value assessment, circular economy principles, advanced technologies, interdisciplinary collaboration, and long-term societal benefits. This study identifies six key research themes in recycled building materials: life cycle assessment, biological and natural materials, recycled concrete, recycled asphalt and building infrastructure, construction and demolition waste, and environmental impacts with composite factors. Furthermore, current research is categorized into two primary dimensions: value strategies and technological tools. The analysis of future research directions underscores the potential of AI-driven innovations and their role in enhancing human living environments. However, developing countries continue to face critical challenges, necessitating further interdisciplinary integration and knowledge exchange. Finally, this study proposes a comprehensive and systematic disciplinary framework that offers valuable insights for future strategic planning and technological advancements in the field.

Keywords: recycled materials; construction; life cycle assessment; circular economy; recycled concrete; bibliometric analysis; research trends



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1. Introduction

Recycled materials refer to waste and resource by-products that have been recovered, reprocessed, and repurposed for construction applications. These include construction and demolition (C&D) waste, recycled asphalt, fly ash (FA), agricultural wastes (AWs), waste rubber, and biochar [1–5]. Through advanced processing technologies, these materials are transformed into essential components of new construction projects. The utilization of recycled materials not only mitigates the accumulation of construction waste but also reduces the demand for natural resources and lowers greenhouse gas emissions during the construction process. As a result, the promotion and adoption of recycled materials have become critical pathways toward achieving a sustainable, environmentally friendly, and low-carbon construction industry on a global scale [6].

There has long been a broad consensus in the construction industry that building materials exert a more profound and lasting impact on future climate conditions, social

ethics, and the well-being of future generations than short-term factors such as architectural style and aesthetics [7]. While many contemporary buildings are classified as sustainable, most primarily emphasize energy efficiency during the operational phase, often overlooking the depletion of natural, non-renewable resources associated with the materials used in their construction [8]. Notably, throughout a building's life cycle, raw materials alone account for approximately half of its total carbon emissions [9,10]. According to the United Nations Environment Programme (UNEP), the *Global Status Report for Buildings and Construction* (March 2024) highlights that by the end of 2022, although direct carbon dioxide emissions from building operations had declined, greenhouse gas (GHG) emissions from the production and processing of building materials still constituted nearly 10% of total global emissions. This proportion would be even higher if other building materials, such as plastics, foams, and fabrics, were taken into account [11].

In response to these challenges, the global construction industry has implemented several effective measures in raw material production and construction, including the recycling of various waste materials to reduce natural resource consumption and the monitoring and control of carbon embedded in raw materials [12–15]. However, the advancement and dissemination of technological innovations still rely on a comprehensive understanding of the current value and future trends of recycled materials in the construction industry [16,17]. Existing reviews, however, often adopt a narrow focus on specific technological aspects of recycled materials. For instance, studies frequently emphasize glass, concrete, and plastics derived from C&D waste, while waste wood and insulation materials receive comparatively less attention due to limited technological diffusion and lower profit margins. This fragmented approach fails to address the broader governance and cognitive challenges associated with recycled materials, despite their significant contribution to building-related carbon emissions [18,19]. Moreover, such limitations obscure the intricate interplay between policy, societal factors, and the well-being of future generations in the context of construction materials. Few studies establish connections between construction firms and waste recycling businesses to assess the effectiveness of recycled material applications, overlooking the construction industry's complexity as a multidisciplinary and cross-sectoral systems project [20–22]. Although Chen et al. provide a relatively comprehensive review of the field, their study offers only a limited analysis of six organic materials, including biochar, bioplastics, Aws, and wool, while broadly categorizing technological advancements in nanocomposites, additive manufacturing, and related fields simply under concrete. Additionally, their analysis lacks significant time sensitivity [23,24]. In addition, research on recycled building materials has advanced rapidly with recent developments in artificial intelligence (AI) technology. The recycling, research, and reuse of traditional materials are being transformed by more efficient machine learning (ML) methods [25,26]. However, a systematic literature review is lacking to integrate these significant global technological advancements with comprehensive research on recycled building materials. The broader literature also tends to focus either on technological advancements or policy considerations, further underscoring a critical gap in current research on this subject.

Thus, this study aims to present a comprehensive and systematic review of the application of recycled materials in construction. Firstly, an extensive literature review was conducted using two bibliometric analysis software programs to generate a knowledge map, which includes outputs such as authorship networks, country and institutional distributions, and keyword co-occurrence analysis, thereby capturing the overall research landscape in the field. Secondly, research trends and emerging developments in recent years were examined to forecast the future trajectory of the field. By providing a comprehensive, systematic, and forward-looking review, this study serves as a valuable reference

for both current and future applications of recycled materials in construction, as well as for policymaking [27].

2. Data Sources and Methods

2.1. Data Source

This study adopts a comprehensive strategy for literature data collection and analysis. Firstly, the Web of Science Core Collection (WoS CC) was selected as the primary data source, as it is widely recognized as a high-quality literature database. The WoS CC contains articles with higher citation rates than many other databases [28], and its structured data format is particularly suited for complex citation network analysis. Thus, it is regarded as the most suitable data source for bibliometric analyses [29–31]. Secondly, previous bibliometric studies have shown that research papers and reviews hold greater value than conference proceedings and book reviews in such analyses [32,33]. In summary, the literature data collection for this study is presented in Table 1.

Table 1. Summary of data source and selection.

Category	Specific Standard Requirements
Research database	Web of Science Core Collection
Citation indexes	All
Language	English
Query preview	((TS = ("recycled materials" OR "sustainable materials") And TS = ("construction" OR "architecture" OR "building") And LA = ("English")) AND DT = ("Article" OR "Review"))
Document types	Article, Review Article
Data extraction	Full Record and Cited References
Sample size	1791

2.2. Data Cleaning

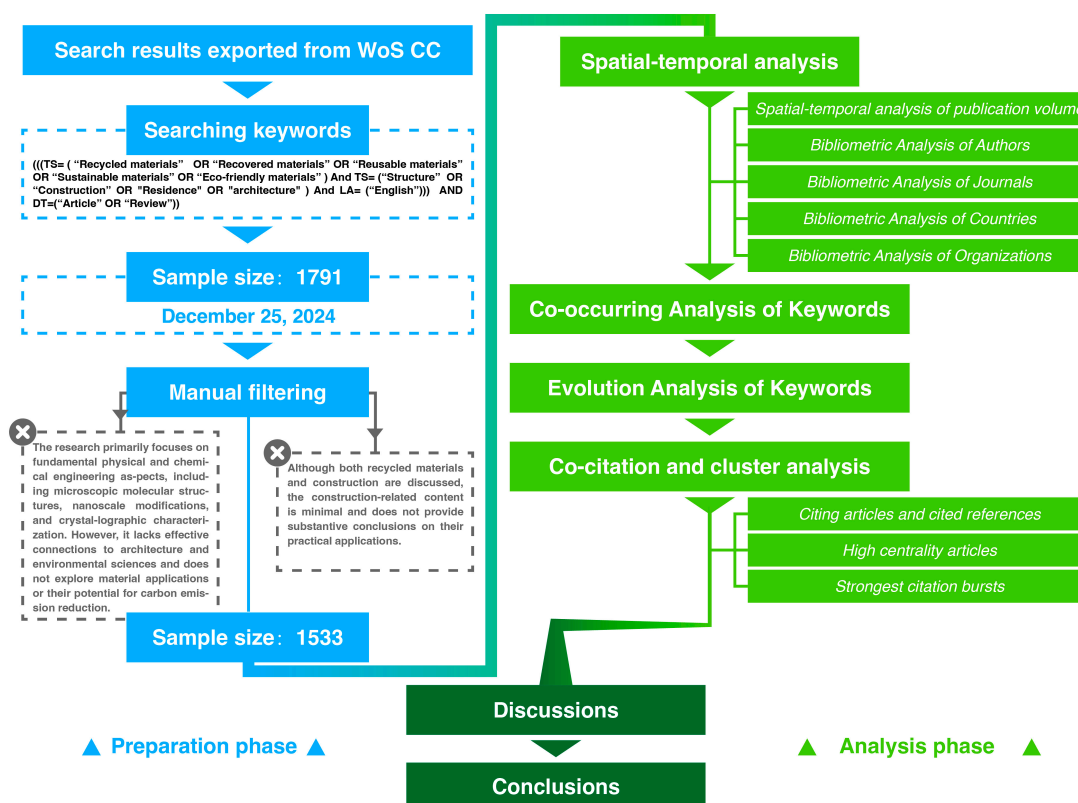
Following the direct export of resources from the WoS CC database, 1791 literature records meeting the search criteria were obtained. Subsequently, all titles and abstracts were carefully reviewed to exclude studies that were irrelevant to the research topic or unsuitable for inclusion in this study, based on the following two criteria:

- The research primarily focuses on fundamental physical and chemical engineering aspects, including microscopic molecular structures, nanoscale modifications, and crystallographic characterization. However, it lacks effective connections to architecture and environmental sciences and does not explore material applications or their potential for carbon emission reduction.
- Although both recycled materials and construction are discussed, the construction-related content is minimal and does not provide substantive conclusions on their practical applications.

After manual screening and deduplication, 1533 valid literature records remained, comprising 1326 research papers and 207 review papers. These records span 439 journals, 5601 authors, 1850 institutions, and 104 countries and regions, incorporating 76,772 references from 25,541 journals. The specific statistical results are presented in Table 2, while Figure 1 illustrates the process of literature acquisition, screening, and subsequent analysis.

Table 2. Data summary after cleaning.

Category	Data
Total	1533
Article	1326
Review Article	207
Journals	439
Authors	5601
Organizations	1850
Countries/Regions	104
References	76,772
Cited Journal	25,541

**Figure 1.** Research process.

2.3. Research Methods

Bibliometrics is a crucial tool for assessing research output, inter-institutional collaboration, and national influence [34,35]. In specific research contexts, bibliometric analysis is typically conducted in two stages: performance analysis and knowledge mapping [36]. Accordingly, this study first conducts a performance analysis based on the cleaned literature data to identify publication trends, key authors, leading institutions, and national contributions in the field. Secondly, to further examine shifts in research themes and key findings, clear and effective visualization maps were generated using two widely recognized bibliometric software programs, VOSviewer (version 1.8.0_441) and CiteSpace (version 6.4.R1) [37–41]. Finally, through multivariate data integration and analysis, this study predicts future prospects and trends in the application of recycled materials in construction.

3. Results

3.1. Spatial–Temporal Analysis

3.1.1. Spatial–Temporal Analysis of Publication Volume

Based on the analysis and categorization of the time span of the existing body of literature, the application of recycled materials in construction has been divided into three developmental stages, as illustrated in Figure 2.

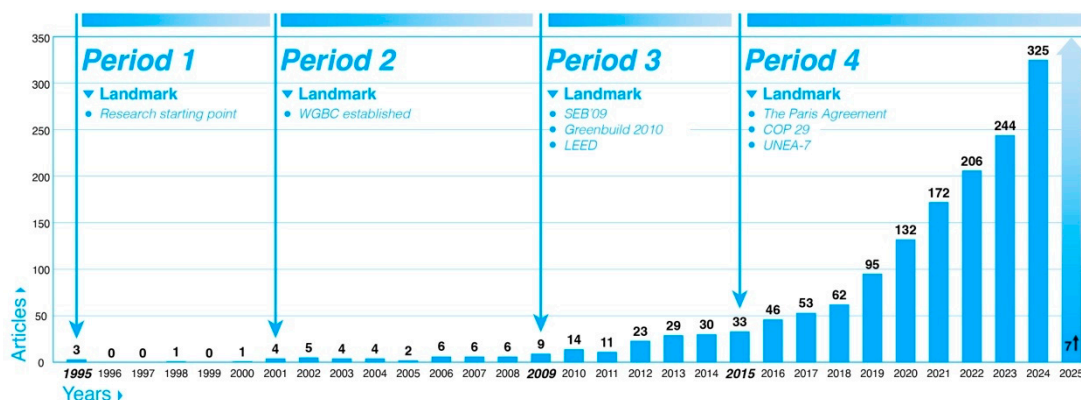


Figure 2. Publication volume analysis.

Phase 1 (1995–2002): This phase marked the initial stage of development for recycled materials, characterized by limited research output. During this period, the construction industry focused on addressing the significant housing demand driven by rapid global population growth [42], and the emerging research hotspots included the structural reliability of building materials and the efficient use of energy in buildings [43,44]. The comprehensive exploration of building materials during this time laid a critical foundation for the subsequent emergence and promotion of recycled materials [45,46].

Phase 2 (2002–2009): The establishment of the World Green Building Council (WGBC) in 2002 was a pivotal event in this period. At its inaugural meeting, the WGBC highlighted the significant economic benefits that green buildings would offer, emphasized the importance of global collaboration in this field, and initiated the promotion of green building certification systems [47]. However, during this period, with the exception of a few countries that began to focus on the substantial resource consumption and carbon emissions associated with building materials, most regions, such as Asia and South America, continued to prioritize addressing the significant housing shortage and focused primarily on the development of new building projects and the immediate economic benefits offered by the traditional construction industry. As a result, the literature on this field remained inconsistent. Nevertheless, these developments marked the beginning of systematic and organized research into recycled materials and green buildings.

Phase 3 (2009–2015): In 2009, the first International Conference on Sustainable Energy and Buildings (SEB'09) presented the latest research findings in the field, including low-carbon buildings based on intelligent software agents, the potential of straw as an exterior cladding material, and the exploration of the physical properties of straw bales [48,49]. As a result, the potential of building materials in terms of recycling and emission reduction gained international recognition [50,51]. Since then, the annual average number of publications has steadily increased to over ten. The subsequent Green Building International Conference and Expo, held in 2010, also highlighted the potential of recycled aluminum in building applications, establishing it as one of the earliest globally recognized recycled building materials of significant importance [52]. Notably, the LEED (Leadership in Energy and Environmental Design) rating system developed by the U.S. Green Building Council

(USGBC) gained widespread adoption during this period, and C&D waste management, raw material procurement, and composition were incorporated into its indicator system, making it an authoritative index for green building development and certification [53,54].

Phase 4 (2015–present): The signing of the Paris Agreement marked the beginning of a period of rapid development in recycled building materials. During this phase, the number of research publications has increased significantly, and various new materials, such as waste rubber and recycled asphalt, have been extensively explored and successfully applied [55,56]. Additionally, materials like recycled concrete and bio-waste, which have been under development for over a decade, now constitute a substantial proportion of construction materials, making a significant contribution to reducing carbon emissions from buildings [57]. However, it is crucial to recognize that a considerable body of literature and conference reports indicates that the industry is currently at a critical juncture, where both risks and opportunities coexist. Moreover, there remains a substantial gap in meeting the global target of reducing carbon emissions by 2030 [58]. Although innovative materials such as bamboo, hemp fiber, and even fungal-based materials continue to emerge, their actual performance and application potential have yet to be fully validated [59]. Furthermore, the intricate relationships between materials, policies, and taxation remain unclear in many countries, highlighting the urgent need for further research and exploration in this field [60,61].

3.1.2. Bibliometric Analysis of Authors

A survey of author groups in the current research can provide insights into the most influential and pioneering scholars in the field. Based on VOSviewer’s analysis of author publications and citation counts, there are seven authors with at least seven publications and nineteen authors with six or more. To enhance the efficiency and focus of this study, only authors with seven or more publications are presented in the author profile, ranked according to their average citations per article (Table 3).

Table 3. Publications by key authors.

Rank	Author	Documents	Citations	Average Citation/Publication
1	José Ramón Jiménez	7	360	51.43
2	Jesus Ayuso	9	438	48.67
3	Francisco Agrela	16	609	38.06
4	Clara Celauro	7	239	34.14
5	Cesare Sangiorgi	7	225	32.14
6	Suksun Horpibulsuk	20	515	25.75
7	Arul Arulrajah	25	624	24.96

According to the statistical results and specific literature analysis, the three most influential authors in this field are Jiménez, Ayuso, and Agrela. These researchers frequently collaborate and are affiliated with the Department of Architectural Engineering at the University of Cordoba, with their research primarily focused on the aggregate composition and properties of recycled concrete [62,63]. Arulrajah and Horpibulsuk, who have the highest number of publications, are from Australia and Thailand, respectively, and have also established a strong collaborative relationship. Their research focuses on composite recycled materials, including discarded rubber, fly ash, and glass. They have maintained a high level of research activity in recent years, exemplifying successful transnational cooperation in this field [64,65]. Notably, the research of nearly all high-impact authors extends beyond the application of recycled materials in buildings to areas such as road transportation and infrastructure [66,67], highlighting the widespread potential of recycled

materials and the increasing recognition of interdisciplinary collaboration within the global academic community.

3.1.3. Bibliometric Analysis of Journals

An analysis of the top ten journals with the highest number of publications in this field reveals that the majority belong to the domain of architecture and building technology, with only a few classified under environmental sciences, as shown in Table 4.

Table 4. Core journal analysis.

Rank	Source	Publications	Citations	Average Citation/Publication
1	Construction and Building Materials	144	5133	35.65
2	Sustainability	127	1379	10.86
3	Journal of Cleaner Production	75	3598	47.97
4	Materials	72	1077	14.96
5	Journal of Building Engineering	53	1368	25.81
6	Transportation Research Record	36	523	14.53
7	Buildings	34	253	7.44
8	Resources Conservation and Recycling	26	2373	91.27
9	Journal of Materials in Civil Engineering	26	1333	51.27
10	Case Studies in Construction Materials	26	355	13.65

Among the top ten journals with the highest number of publications, seven support open access for all or most of their articles, including *Construction and Building Materials*, *Journal of Cleaner Production*, and *Materials*. This trend suggests that open access has significantly contributed to the advancement of the field in recent years. Additionally, statistical analysis indicates that *Resources, Conservation and Recycling* ranks as the top journal in this domain, with an average of 91.27 citations per article, highlighting the high quality of its publications. A review of articles on recycled materials published in the past three years reveals that this journal emphasizes the comprehensive application of recycled materials in construction, supported by empirical research [68]. Furthermore, it explores the economic, policy, and broader social dimensions of material use in case studies [69,70], reflecting the prevailing research trends among the most influential works in this field.

3.1.4. Bibliometric Analysis of Countries

To identify the countries that have made significant contributions to the field of recycled building materials, an analysis was conducted on the number of publications and international collaborations. Figure 3 illustrates the contributions of 57 countries, each with more than five publications.

In Figure 3, the larger the node labeled with a country's name, the greater the number of publications from that country in this field. The thickness of the connecting lines between nodes represents the intensity of collaboration between two countries. The chart indicates a relatively balanced distribution of contributions across countries. Further analysis reveals that the United States and Australia are the most active in international collaboration, exhibiting the highest node connectivity strength (exceeding 500). Additionally, Table 5 presents the top five countries in terms of global publications. The results indicate that the United States holds the most influential and representative position in this field, as evidenced by its highest number of publications and citations per article.

Counting Method: Full Counting

Full counting means that each co-authorship, co-occurrence, bibliographic coupling, or co-citation link has the same weight.

Threshold number of publications: 5

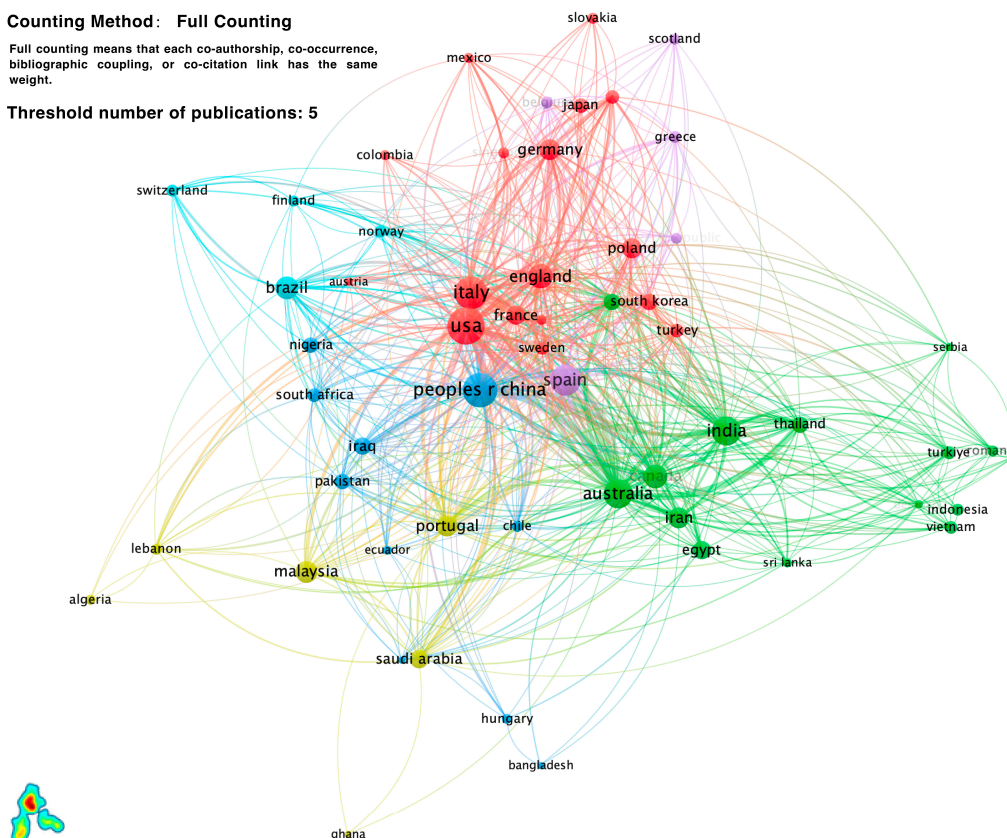


Figure 3. Co-occurrence of countries.

Table 5. Top 5 countries.

Rank	Source	Documents	Citations	Average Citation/Publication
1	USA	216	8364	38.72
2	China	176	4711	26.77
3	Italy	158	4251	26.90
4	Spain	134	2711	20.23
5	Australia	121	3901	32.24

3.1.5. Bibliometric Analysis of Organizations

Figure 4 illustrates the current distribution of institutional and organizational publications. The analysis indicates that Swinburne University of Technology has the highest number of global publications and the most extensive research collaboration network. A closer examination of its research output reveals a primary focus on infrastructure development and the advancement of regenerative materials [71–73], along with a strong collaborative relationship with Suranaree University of Technology in Thailand. However, the number of publications alone does not fully reflect an institution's influence in this field. According to an analysis of the top five institutions by publication volume (Table 6), the Royal Melbourne Institute of Technology University emerges as a key research hub, with an average of 46.20 citations per article, underscoring its significant academic impact. Furthermore, the findings highlight the critical role of universities and research institutions worldwide in advancing research on recycled building materials.

Counting Method: Full Counting

Full counting means that each co-authorship, co-occurrence, bibliographic coupling, or co-citation link has the same weight.

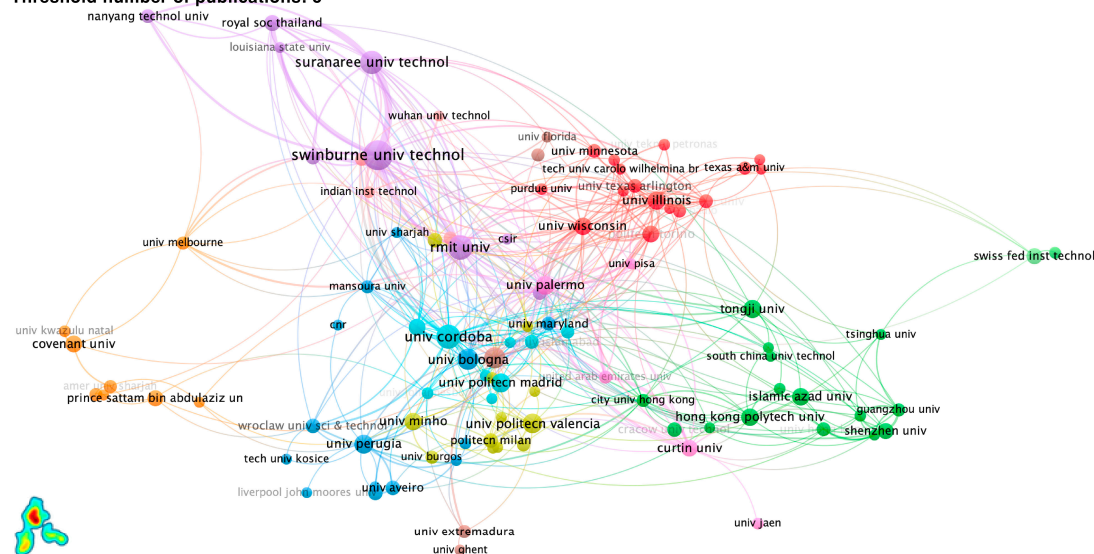
Threshold number of publications: 5

Figure 4. Co-occurrence of organizations.

Table 6. Top 5 organizations.

Rank	Organization	Documents	Citations	Average Citation/Publication
1	Swinburne University of Technology	33	1354	41.03
2	Royal Melbourne Institute of Technology University	24	1109	46.20
3	University of Córdoba	22	777	35.32
4	Suranaree University of Technology	21	562	26.76
5	University of Lisbon	17	538	31.65

3.2. Keywords Cluster Analysis

The visual analysis of keywords identified six thematic clusters, with the graphical representation demonstrating a comprehensive and well-balanced distribution. The colors corresponding to each cluster are labeled in the upper-right corner of Figure 5, ensuring clear differentiation and interpretation.

Among the 6346 keywords, 468 with at least five occurrences were selected for cluster analysis. This threshold was established to maintain the focus of the research topic and prevent redundancy caused by excessive keywords. Clustering is primarily based on keyword correlations. A more distinct clustering color block indicates stronger keyword connections within a category, reflecting greater thematic convergence. Clusters are ranked according to the number of keywords and the degree of integration.

The results indicate that Cluster 1 (red) focuses on the application strategies and value assessment theories of recycled building materials, including life cycle assessment (LCA), the sustainability of green buildings, energy efficiency, and carbon emission management in the industry. Cluster 2 (green) centers on the processing technologies of biological and natural materials, including biofibers, biochar, AWs, and nanocomposites. Some studies have also explored the structural strength and acoustic properties of biomaterials. Cluster 3 (blue) is concerned with cement and recycled concrete, specifically the composition of aggregates and the structural properties of concrete, making it one of the most widely used recycled building materials. Cluster 4 (yellow) focuses on recycled asphalt and road engineering, infrastructure, and related areas. The significant overlap of this theme with

other clusters suggests a trend of cross-disciplinary research in construction and civil engineering. Cluster 5 (purple) addresses the recycling and performance testing of C&D wastes, such as crushed glass, mortar, slag, and elastic modulus testing of these materials. Cluster 6 (blue-green) represents a composite theme involving economic feasibility analysis, engineering characteristics, and soil environmental changes, linking several of the aforementioned research dimensions.

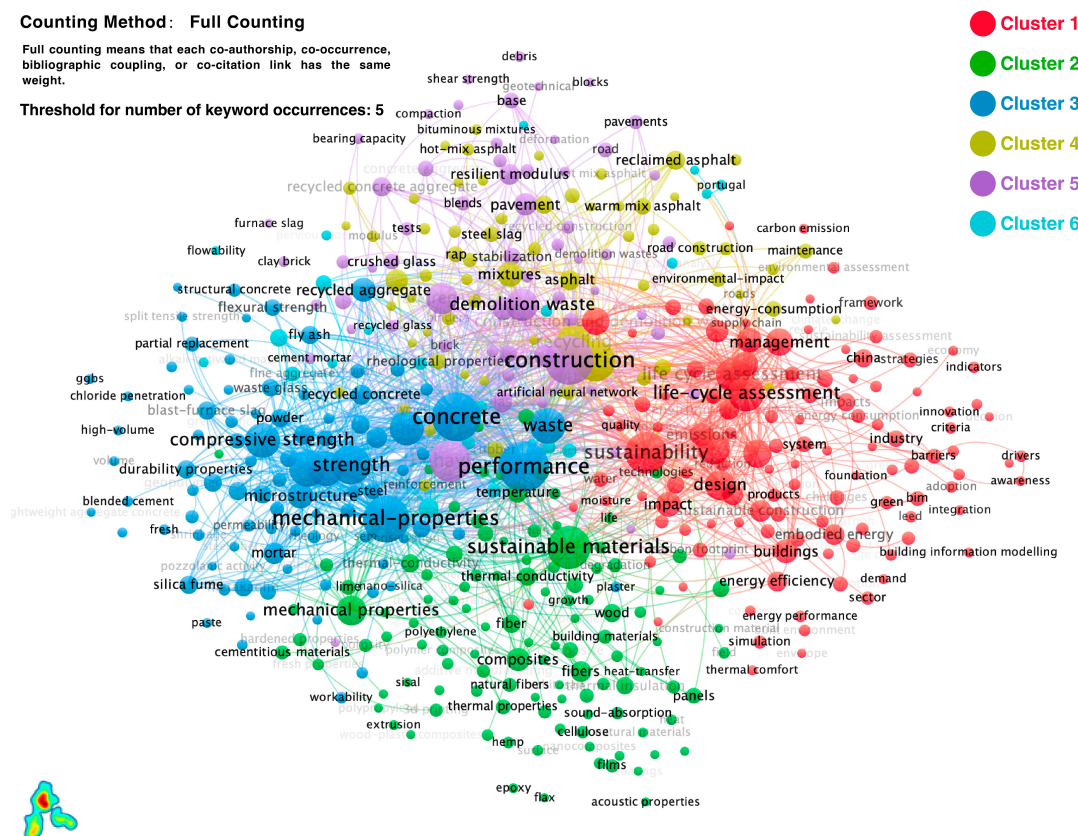


Figure 5. Keyword cluster analysis.

The clustering results effectively capture the current research landscape of recycled building materials. Clusters were ranked based on keyword count and integration level. The first five clusters exhibit distinct themes, whereas Cluster 6 is more diverse, overlapping with other themes and demonstrating a tendency for deeper integration.

3.3. Evolution Analysis of Keywords

Analyzing the spatial and temporal evolution of keywords based on their clustering helps to understand the macro-level evolution and development of research topics on recycled building materials. Figure 6 illustrates the distribution of keywords over time, generated using CiteSpace.

The clustering results indicate that the concept of recycled materials was widely used in academic publications starting in 2001, followed by less than a decade of research focusing on carbon implications in materials and LCA. Keywords such as concrete and recycled asphalt emerged around 2010, laying the foundation for subsequent studies on recycled aggregates and C&D waste management. The number of keywords increased rapidly after 2015, and their prominence in recent years suggests significant advancements in areas such as additive manufacturing, C&D waste management, and the circular economy (CE).

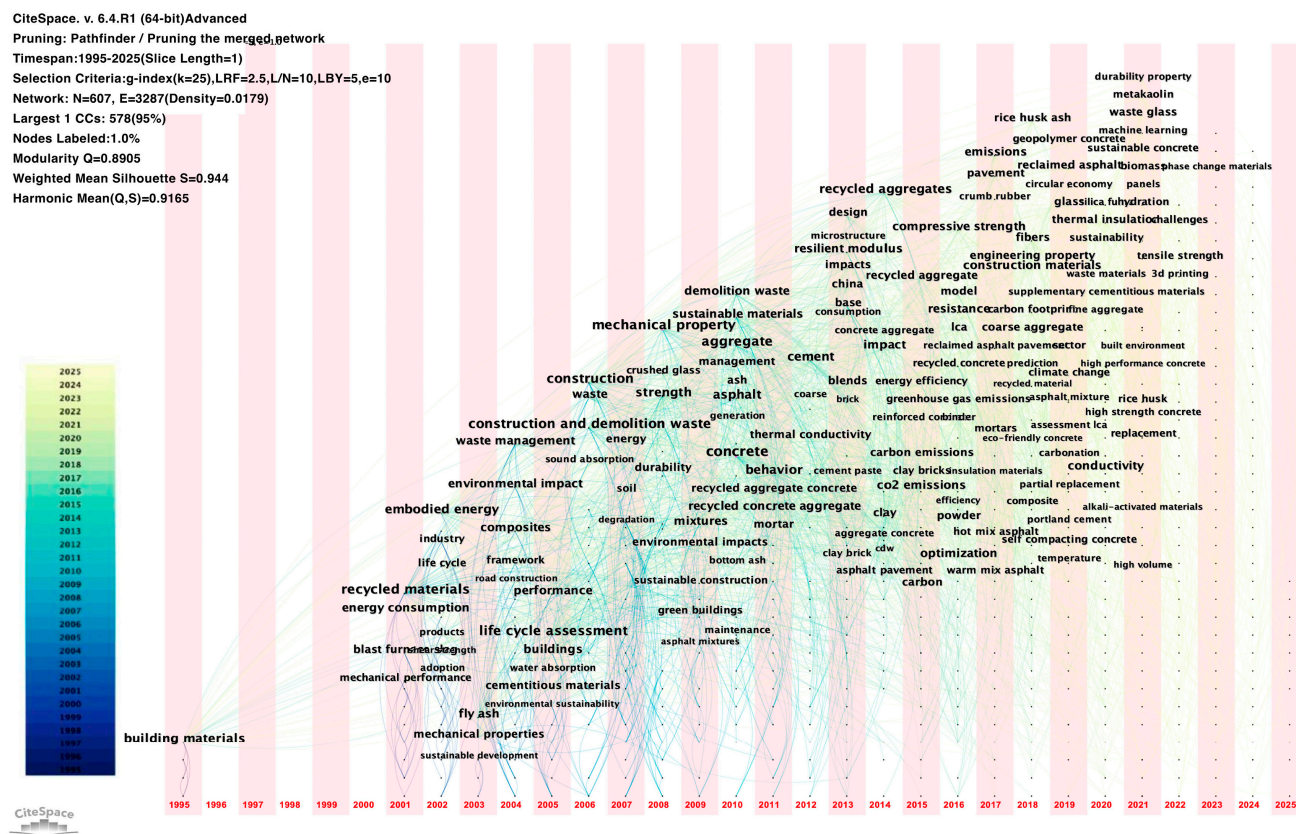


Figure 6. Map of timezone view.

3.4. Citation Network Analysis

3.4.1. Citing Articles and Cited References

Cluster analysis of the cited literature was conducted using CiteSpace, and the modularity value of the clustering results ($Q = 0.8905$) along with the weighted average silhouette ($S = 0.944$) indicate that the clustering results are significant and highly reliable (Figure 7). A total of 11 valid clusters currently exist, with the background color change of the clusters reflecting the difference in the year of their appearance. It is evident that most of the clusters began to emerge in the last decade. Table 7 provides the details of each cluster in descending order of influence.

Figure 8 shows the temporal evolution of the citation clustering results, with the size of the nodes indicating the frequency of citations, and the position of the different topics in the timeline representing their level of interest across different time periods. Three clusters—Cluster #3 (demolition material), Cluster #11 (evolutionary properties), and Cluster #12 (fine recycled concrete aggregate)—demonstrate a significant decline in citations in recent years, suggesting that research in these areas may have become outdated or merged with other fields. On the other hand, Clusters #0 (sustainable construction), #1 (life cycle assessment), #4 (economic analysis), and #7 (bearing mixed recycled aggregate) have not only been active for a long time but have also maintained a high level of citations in recent years, highlighting the core areas and research hotspots in this field.

The top three influential clusters in the co-citation analysis are now specifically analyzed, along with a review of the titles and abstracts of the top five articles in terms of citations for each cluster. This analysis aims to uncover the complex interconnections between the various research areas on recycled building materials and their corresponding key findings. The most cited literature in Clusters #0, #1, and #2 are presented in Tables 8–10, respectively.

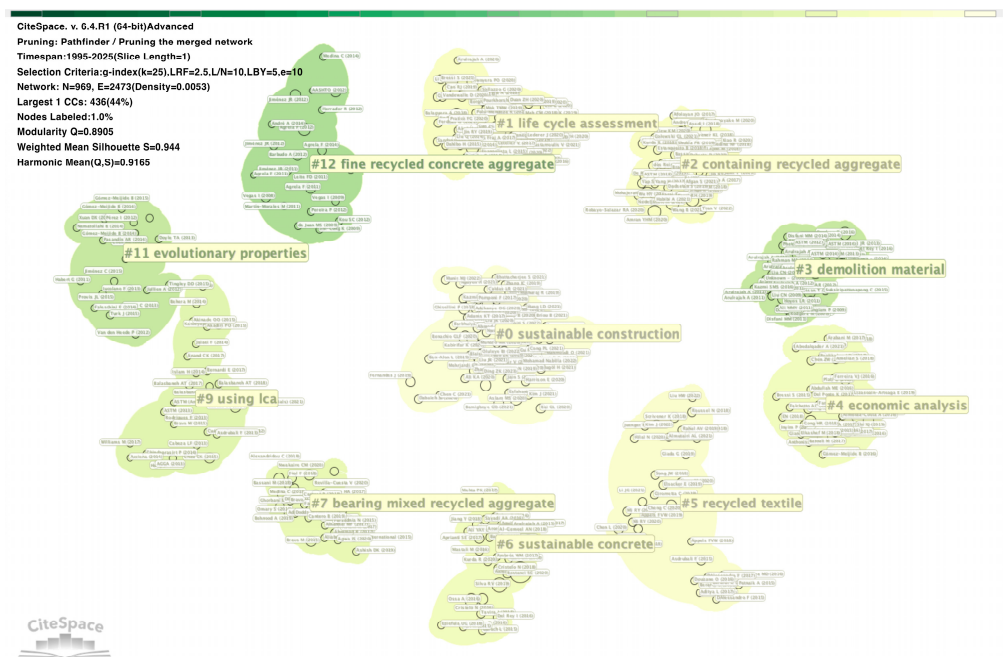


Figure 7. Co-citation cluster analysis of the literature.

Table 7. Co-citation clustering result details.

Cluster ID	Size	Silhouette	Mean (Year)	Label
0	74	0.97	2020	Sustainable construction
1	60	0.841	2018	Life cycle assessment
2	53	0.933	2019	Containing recycled aggregate
3	45	0.998	2012	Demolition material
4	37	0.925	2017	Economic analysis
5	34	0.959	2018	Recycled textile
6	33	0.929	2017	Sustainable concrete
7	29	0.961	2017	Bearing mixed recycled aggregate
9	27	0.997	2015	Using lca
11	22	0.994	2011	Evolutionary properties
12	22	0.946	2013	Fine recycled concrete

Table 8. Citation and the cited literature for Cluster #0.

Citing Articles in Cluster #0		Cited References in Cluster #0	
Author (Year)	Coverage	Author (Year)	Freq
Firoozi, et al. [74] (2024)	22%	Benachio, et al. [75] (2020)	9
Liu, et al. [76] (2024)	8%	Ghisellini, et al. [77] (2018)	8
Martin, et al. [78] (2024)	8%	Aslam, et al. [79] (2020)	7
HaitherAli and Anjali [80] (2024)	7%	Kabirifar, et al. [81] (2020)	6
Alazaiza, et al. [82] (2024)	7%	Han, et al. [83] (2021)	6

Cluster #0 has the highest impact to date, focusing on the strategic level of recycled building materials. In addition to the main tags, other core terms include demolition waste, CE strategy, comparative analysis, and climate change, bringing together the latest results from current strategy-level research, such as material inventories, LCA, and building materials passports (BMPs), among others. Material inventories rely on the analysis of current building profiles in cities to capture the stock of materials available for demolition and recycling at the end of a building's life cycle, with current outputs including several GIS assessment models and databases [84–86]. Meanwhile, LCA analyses focus on the

economic and emission costs of collecting, constructing, and recycling materials, providing important references for analyzing the consumption and utilization rates of current recycled materials [87]. BMPs are designed to create a profile of materials and structures at the beginning of the building construction phase, to evaluate the recyclable components of the building during the demolition phase [88,89].



Figure 8. Co-citation clustering evolution.

Table 9. Citation and the cited literature for Cluster #1.

Citing Articles in Cluster #1		Cited References in Cluster #1	
Author (Year)	Coverage	Author (Year)	Freq
Zhang, et al. [90] (2019)	12%	Tam, et al. [91] (2018)	25
Gherman, et al. [92] (2023)	9%	Gálvez-Martos, Styles, Schoenberger and Zeschmar-Lahl [18] (2018)	21
Kolaventi, et al. [93] (2022)	7%	Balaguera, et al. [94] (2018)	11
Illankoon and Vithanage [95] (2023)	7%	Huang, et al. [96] (2018)	11
Bayram and Greiff [97] (2023)	7%	Borghi, et al. [98] (2018)	10

Cluster #1 exhibits the longest life cycle and primarily focuses on the recycling of LCA and C&D waste. Other core terms include mixed recycled aggregates, mortar materials, and waste concrete. The global demand for aggregates currently reaches an astonishing 40 billion tons per year, highlighting the ongoing intensity of construction activities in most regions [99]. Meanwhile, the shortage of landfill space and the challenges of waste recycling suggest a looming crisis in this field, further exacerbated by legal deficiencies in many developing countries, which hinder the effective utilization of C&D waste. The findings in Cluster #1 reflect ongoing research advancements in policy development, global warming potential (GWP), and soil environmental changes. Given the critical nature of these issues, research within this cluster is expected to remain central to the field of recycled building materials for the foreseeable future [97].

Table 10. Citation and the cited literature for Cluster #2.

Citing Articles in Cluster #2		Cited References in Cluster #2	
Author (Year)	Coverage	Author (Year)	Freq
Mei, Xu, Ahmad, Khan, Amin, Aslam and Alaskar [3] (2022)	8%	Mohajerani, Burnett, Smith, Markovski, Rodwell, Rahman, Kurmus, Mirzababaei, Arulrajah and Horpibulsuk [4] (2022)	14
Nalon, et al. [100] (2022)	7%	Nedeljković, et al. [101] (2021)	11
Valente, et al. [102] (2022)	7%	Amran, et al. [103] (2020)	10
Alhawati, et al. [104] (2022)	6%	Wang, et al. [105] (2021)	10
Kim, et al. [106] (2022)	6%	Liew and Akbar [107] (2020)	7

In contrast to the macro-policy and economic cost considerations emphasized in the first two clusters, Cluster #2 focuses on the specific technical aspects of recycled materials. Core terms include mechanical properties, polyethylene terephthalate (PET) aggregates, and alkaline activation. Key research areas in this cluster involve the physical properties of fine recycled concrete aggregate (fRCA), PET aggregates, and geopolymer concrete (GeoPC). Additionally, Cluster #2 exhibits a strong correlation with Cluster #5 and Cluster #6, further highlighting its focus on material development technologies. Although the number of publications in Cluster #12 remains relatively low, its keywords still appear prominently in Cluster #2, indicating that the application potential of fRCA has been extensively discussed and its reliability validated, making it a crucial component of modern recycled building materials.

3.4.2. High-Centrality Articles

This section examines the contribution of the top five articles with the highest centrality (Figure 9). The centrality of an article within the citation network directly reflects its influence on the research field. A higher centrality indicates that the study integrates and impacts a broader range of related research, addressing multiple complex issues at the intersection of various topics. Consequently, such articles play a pivotal role in shaping the development and depth of research in this field (Table 11).

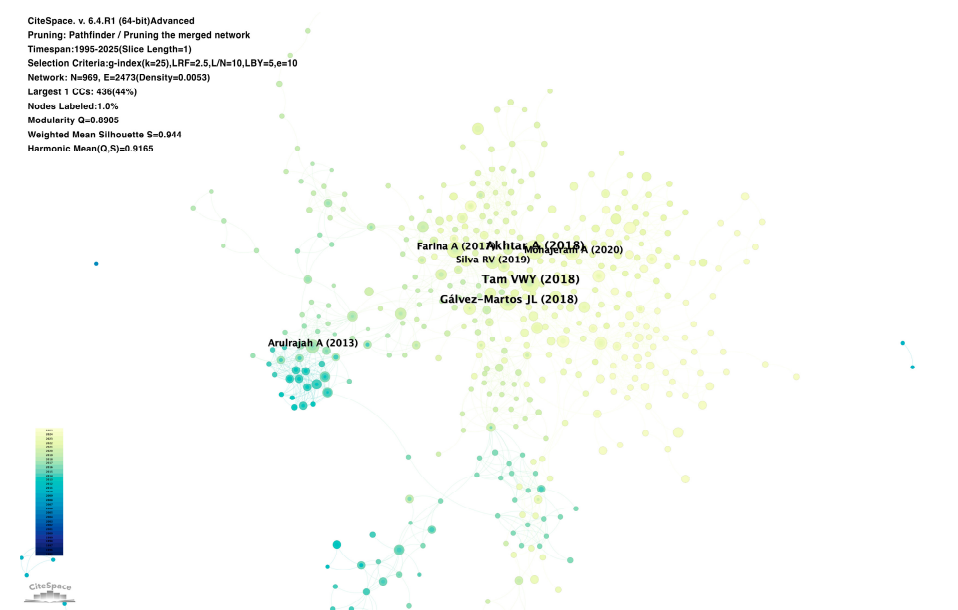
**Figure 9.** Cited literature centrality analysis (see Appendix A).

Table 11. Detailed explanation of the top five high-centrality articles.

Year	Centrality	Articles	Contributions
2020	0.07	Benachio, Freitas and Tavares [75]	- Identifies the current gap in understanding of CE practices within the construction industry. - Argues that the economic operational model of construction companies plays a more crucial role in establishing a circular economy system than extensive policy formulation and publicity.
2015	0.07	Bravo, et al. [108]	- Analyzes the mechanical properties of C&D waste and coarse recycled aggregate. - Examines the effects of source, composition, particle size, and blending ratio on RA performance.
2014	0.05	Arulrajah, et al. [109]	- Conducts shear, triaxial, and unconfined compression strength tests on various C&D materials. - Highlights the broad applicability of C&D materials in construction.
2018	0.05	Balaguera, Carvajal, Albertí and Fullana-i-Palmer [94]	- Literature review on LCA applications in road engineering. - Identifies the need for developing countries to prioritize road infrastructure designed for lower traffic volumes.
2018	0.05	Huang, Wang, Kua, Geng, Bleischwitz and Ren [96]	- Analyzes the current state of C&D waste management in China. - Proposes strategic solutions to enhance C&D waste management in China based on existing challenges.

3.4.3. Strongest Citation Bursts

The term “citation burst” refers to a sudden surge in the number of citations to a particular article within a specific period. Analyzing these articles helps to effectively identify research hotspots over time and predict future trends. Figure 10 illustrates citation bursts from 1995 to the present, with the majority being review papers. This predominance is due to the ability of review articles to synthesize prior research findings in a given field and provide valuable insights for subsequent studies. Table 12 presents an analysis of the contribution levels of the five most recent citation bursts.

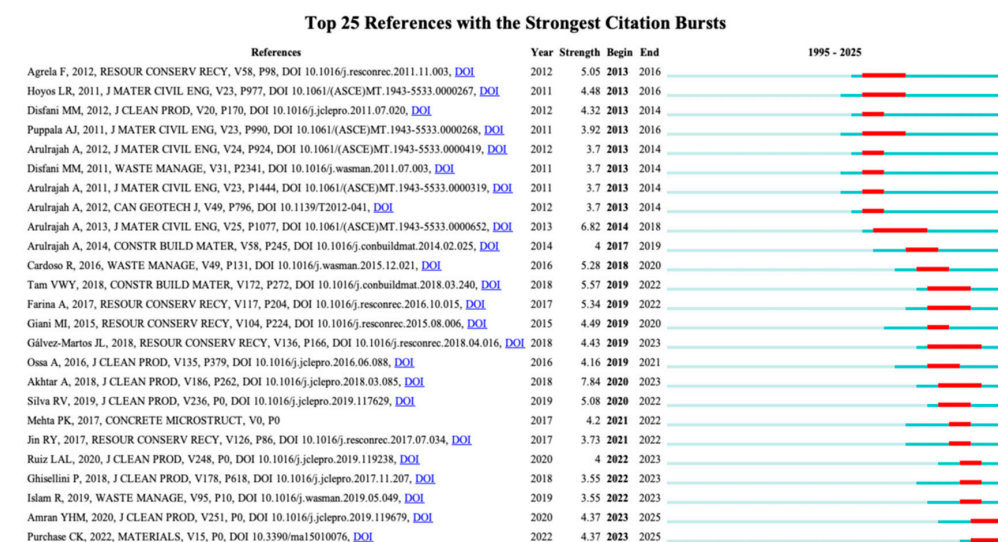
**Figure 10.** Top 25 references with the strongest citation bursts (see Appendix B).

Table 12. Analysis of the contribution of the latest five citation bursts.

Begin	End	Articles	Contributions
2022	2023	Ruiz, et al. [110]	- Proposes key strategies for implementing CE principles in the C&D waste sector, structured into five phases. - Concludes that pre-construction and demolition strategies have the greatest impact on CE operations.
2022	2023	Ghisellini, Ripa and Ulgiati [77]	- Demonstrates the feasibility of the CE framework for future applications. - Analyzes multiple factors influencing the effectiveness of CE framework implementation.
2022	2023	Islam, et al. [111]	- Conducts a systematic investigation of C&D waste generation and performs a comparative analysis of national data. - Emphasizes the importance of disaggregated C&D waste data in developing effective waste management plans.
2023	2025	Amran, Alyousef, Alabduljabbar and El-Zeadani [103]	- Examines the physical properties of GeoPC. - Highlights the role of GeoPC in the future development of sustainable buildings.
2023	2025	Purchase, et al. [112]	- Identifies key barriers to the current implementation of the CE in the C&D waste sector. - Provides a quantitative data reference for establishing a future global CE framework.

4. Analysis of Results

4.1. Research Developments

The number of publications on recycled building materials has steadily increased since 1995, exhibiting an exceptionally high growth rate in recent years. This trend highlights both the significant advancements in research on recycled materials and the growing global awareness of the negative environmental impact of carbon emissions from buildings. The evolution mapping in Figure 6 illustrates that core keywords such as recycled concrete, LCA, and C&D waste—widely used since the early 2000s—have retained their relevance despite the emergence of numerous new keywords in recent years. An analysis of authors, countries, institutions, and journals indicates that the United States and Australia lead in multinational collaborations, ranking highest in both publication volume and citation impact. Among institutions, the Royal Melbourne Institute of Technology University and Swinburne University of Technology—both in Australia—are the most influential. At the author level, José Ramón Jiménez from Universidad de Córdoba, Spain, is identified as the most influential researcher in this field. Different countries and institutions have distinct research priorities. In recent years, China has focused on the application of recycled materials in pavilion design, low-carbon cities, and energy subsidies [113–115], reflecting its progress in green energy, though it still lacks globally influential scholars and institutions. Italy has explored the potential of recycled materials in urban regeneration and heritage restoration [116–118], likely influenced by its abundant cultural heritage resources. Spain’s research is particularly diverse and innovative, covering topics such as the acoustic properties of fruit kernels and the structural strength of plant fiber blocks in buildings [119,120]. These varying research foci across countries, institutions, and scholars collectively con-

tribute to the advancement of the field, underscoring the importance of global collaboration and knowledge sharing for future developments.

4.2. Co-Occurring Analysis of Keywords

Existing studies identify six major keyword clusters in the field of recycled building materials: LCA and sustainability, biological and natural materials, recycled concrete, recycled asphalt and building infrastructure, C&D waste, and environmental impacts and compounding factors. These clusters collectively form the foundation for this study and subsequent analyses.

Firstly, LCA results have become the most important reference for building planning and material application, directly influencing the further development of materials and the adjustment of CE strategies. This is reflected in their general linkage with all other clusters in the mapping. Evolutionary analysis shows that this concept, which became widely used in publications between 2002 and 2005 throughout various stages of the development of recycled building materials, provides a holistic and systematic approach to evaluating the environmental performance of materials, focusing on quantified carbon emissions, energy consumption, and waste generation across the material's entire life cycle—from raw material extraction to manufacture, transportation, construction, use, and disposal [121]. It is the most crucial analytical tool for assessing the various impacts throughout the life cycle of a material, with Figure 11 illustrating a generic life cycle model for construction materials.

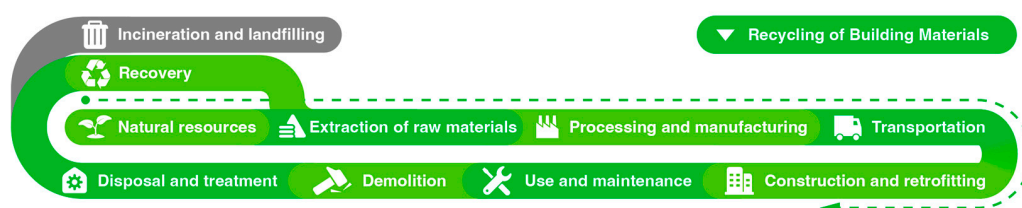


Figure 11. Recycling of building materials.

LCA is categorized into attributional LCA (ALCA) and consequential LCA (CLCA). ALCA assesses the environmental impact of construction materials by tracking inputs and outputs across multiple stages, including raw material acquisition, production, processing, use, and disposal. It employs a static, process-based analysis. In contrast, CLCA extends beyond the product system, incorporating chain reactions and dynamic effects within environmental and socio-economic systems [122,123]. While LCA typically focuses on carbon emissions, there are several precise sub-models within this framework, such as EIO-LCA [124], which incorporates economic inputs and outputs, and uses Location Quotient (LQ) to align with expected economic development goals, enabling more accurate policymaking [120], and Eco-LCA, which quantifies ecological resources and environmental performance, among others [125]. However, the accuracy of LCA results is heavily dependent on the objectivity of data sources and open access to relevant databases. Comparisons of different results require a high degree of consistency in carbon tracking methodologies to achieve the most reliable outcomes [126].

Quantifying uncertainty in the LCA process is crucial for informed decision making. Hoxha, et al. [127] found that while concrete is the largest contributor to a building's carbon emissions, insulation plays a key role in controlling LCA uncertainty, which should be maintained below 20% [128]. Since the environmental impact of building materials depends on site characteristics [129], such as air pollution, land acidification, and particulate emissions [130], regionalizing LCA data is essential for improving reliability [131].

Secondly, biological and natural materials are of great interest due to their easy re-source availability and significant advantages in reducing carbon footprints. They repre-

sent the primary direction of current innovations in renewable building materials, such as biochar, lignocellulose, bioplastics, AWs, and animal hair. Among these, biochar is produced by the thermochemical conversion of biomass from plants and wood at lower temperatures (below 700–800 °C) in an anaerobic environment [132], with sources including, but not limited to, bagasse and rice husk. Typically, the feed concentration of biochar should be kept below 6%, compared to 2% for lignocellulose, which is similar [133]. Studies have shown that biochar and cellulose significantly improve the mechanical properties of concrete, reduce the porosity of cement, and enhance its durability [134,135]. Biochar applications include green roofs with excellent water retention capacity, soil structure stabilization, and carbon emissions reduction, among others, while cellulose can be used to produce biofuels, bioplastics, and other value-added products [136]. In addition, AWs, a relatively comprehensive concept that may include phosphates, nitrogen, organic carbon, and pesticide residues, among others [137], can slow down the degradation process of concrete and improve its flexural and compressive strengths due to the large amounts of potassium and sodium they contain, which react with calcium hydroxide in the concrete to form more stable potassium and sodium compounds [138]. Certain AWs, such as grape pomace and cork, are also made into acoustic panels that achieve an excellent sound absorption coefficient of 0.80 and maintain excellent acoustic performance after repeated use [139]. Figure 12 illustrates the advantages of four biological and natural materials used in construction.

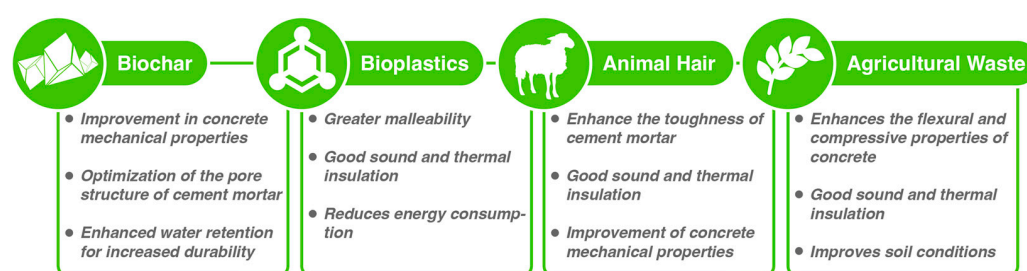


Figure 12. Advantages of four biological and natural materials.

Furthermore, recycled concrete is one of the most widely used recycled building materials, with aggregate alone accounting for 60–75% of its volume [140]. The composition and mechanical properties of recycled concrete aggregate (RCA), among other factors, have attracted extensive attention from scholars. The physical properties of concrete are influenced by multiple factors, such as the original design and recycling and storage techniques. RCA can be sourced from copper tailings and C&D waste, among others [101,141]. The application of RCA not only reduces the demand for natural aggregates (NAs) but also saves land for the disposal of waste concrete. However, due to the high porosity and water absorption of RCA, the compressive strength of concrete is severely reduced when 50% or more of NA is replaced with RCA. The improvement in durability remains the main challenge in the current RCA industry [142]. Therefore, in the production process of RCA, on one hand, water absorption can be reduced by using heated washing (HS) to remove adherent mortar and gravity classification (GC) to separate aggregates according to density, producing different quality levels [143]. On the other hand, a jaw crusher can directly produce aggregates with good profiles, low flakiness, and high productivity from C&D waste, which enhances the compressive strength of RCA with less resource consumption. Although rotary crushers are also widely used to produce RCA, offering higher aggregate quality and lower water absorption, they come with higher energy costs [144]. Finally, methods such as carbonation and acetic acid immersion can increase the compressive strength of RCA to about 87% of that of NA, making it a viable material for various con-

struction purposes [145]. Figure 13 illustrates the composition, use, and recycling process of concrete.

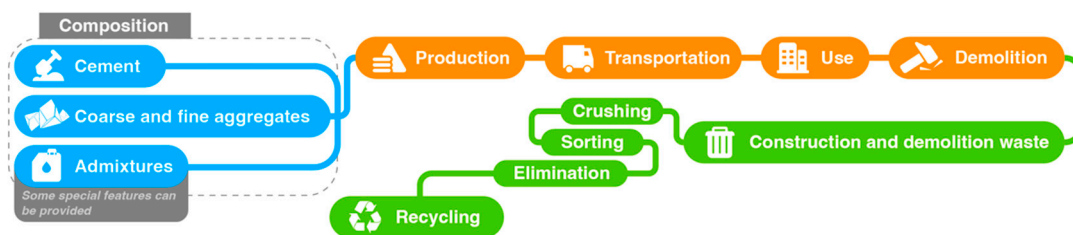


Figure 13. Composition, use, and recycling of concrete.

The last three clusters demonstrate comprehensiveness and a strong correlation with the first three clusters. In the context of construction infrastructure, asphalt is a crucial material for roof waterproofing and building pavement, and Figure 14 illustrates the GHG emissions of asphalt at various stages. However, the traditional hot mix asphalt (HMA) construction method requires temperatures exceeding 375 °F, which significantly increases fuel consumption and GHG emissions. As a result, recycled asphalt has garnered research attention. It can incorporate various types of RCA, such as C&D waste, recycled plastics, and FA, among others, to reduce carbon emissions and natural aggregate (NA) consumption during the construction process. Notably, rubberized asphalt (RA) made from waste tires offers higher cracking resistance and improved fatigue life [146,147]. FA, when used as a filler in HMA, can also be cost-effective by reducing binder consumption, increasing strength, and minimizing physical deformation [148]. Additionally, waste cooking oil (WCO) and waste engine oil (WEO) can serve as asphalt-rejuvenating agents, restoring about 45% of asphalt's original functionality, although volatilization during production remains a pressing issue [149]. Therefore, while new material technologies continue to develop, differences in usage duration and construction sites may alter the properties of reclaimed materials. The systematic material database on which LCA relies is still not established in many developing countries [94], presenting a dilemma for the industry to address in order to facilitate its future development.

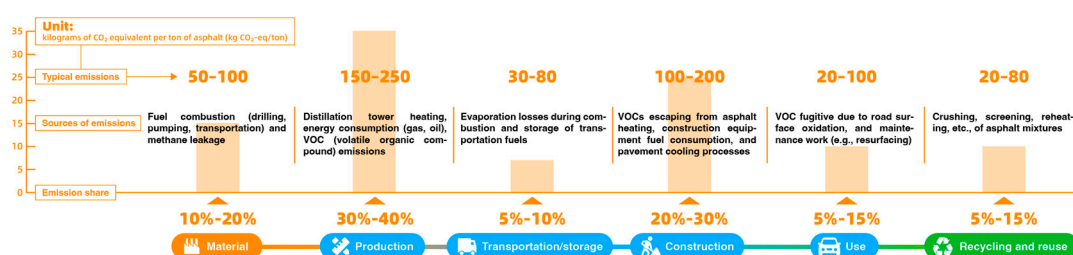


Figure 14. Carbon emissions from asphalt roads.

4.3. Co-Citation and Cluster Analysis

4.3.1. Cluster Analysis

Based on the citation clustering analysis conducted, research related to recycled building materials can be categorized into two levels: value strategies (e.g., Cluster#0, Cluster#1) and technical means (e.g., Cluster#2, Cluster#3). This section will provide an in-depth analysis of the highly cited articles within these two levels to clarify the general connections and overall framework between the different clusters.

Value Strategies of Recycling Building Materials

- The foundation of the research

The analysis of highly cited articles shows that the linear economic model (collect–make–discard), which has long been used in the construction industry, has been wasteful in terms of resources and finances. This has led to the promotion of the CE in recent years by foundations and international organizations, such as the Ellen MacArthur Foundation [75]. The CE is the most important strategic concept in the field of recycled building materials, serving as a sustainable strategy to keep materials within a closed-loop system. This approach maximizes the efficiency of building material usage while reducing the consumption and waste of oil, ore, wood, and C&D waste in the industry. All recycled building materials are key components of the CE [77]. However, the CE is not universally sustainable; it specifically addresses concerns about resource inputs, waste, emissions, and energy leakage [150]. For example, by following CE principles in areas such as material transportation and the construction environment, carbon emissions can be reduced by 50% compared to conventional buildings [151]. Nevertheless, due to the complexity of the CE concept, many decision-makers have a biased understanding, causing its development to be slow [152]. Therefore, some studies have proposed classifying the CE in construction into six dimensions: governmental, economic, environmental, behavioral, social, and technological, in order to further quantify its impact on specific buildings [153]. It has also been suggested that governments should provide incentives to operational and material recycling companies to minimize the impact of conflicts of interest on the CE [154]. However, the CE has also faced criticism in recent years, including concerns about its poor theoretical foundation, lack of structured mentoring programs, and fiscally driven ideologies [155].

The greatest challenge to the CE is the circularity rebound effect (CR), where secondary production of building materials fails to achieve the expected carbon reduction for the industry [156,157]. The CE reduces resource consumption and production costs per unit, yet this may lead governments or builders to increase material use, exacerbating the CR through tax and policy risks. Moreover, the CE remains focused on decision making and consumption rather than replacing primary production, preventing full resource circularity [158,159]. Additionally, material suppliers and builders face “greenwashing” risks, including misrepresentation, partial compliance, or even inaction [160,161]. Achieving the CE requires stronger collaboration among governments, enterprises, and builders, alongside further research on mitigating the rebound effect.

- Research response and progress

Some of the progress of the CE is still worth recognizing. In addition to the previously mentioned material inventories and BMPs, the CE’s emphasis on material selection during the early stages of building design is considered a key initiative to enhance its impact and effectiveness [162]. Currently, a variety of reusability analysis tools for construction waste exist, such as the integration of building information modeling (BIM) with factors like the building’s age to quantitatively assess its renewable material inventory during the early stages of design [163]. It is worth mentioning that LCA, as a core analysis tool for the CE, plays a crucial role in this process. For example, LCA of buildings constructed using Demolishable Design (DfD) revealed that although concrete is the most important component of a DfD building, the benefits of developing a DfD solution for concrete are not significant. In contrast, short-term components such as aluminum profiles, clay tiles, and gypsum wallboard can optimize the environmental effects of the building to a greater extent [164]. However, the current LCA based on CE principles also shows the shortcomings of homogenized indicators. Some studies have incorporated factors such as building type, geographic location, and lifespan, as well as population and urbanization into the LCA through case studies [79], in order to account for the complexity of the building system and the long-term characteristics of buildings [165]. Regarding C&D waste, several studies

have focused on differences in construction and demolition waste management (C&DWM) strategies between countries. The results indicate that in developing countries, C&DWM is often viewed as the responsibility of the government, requiring policymakers to fully understand the CE and the various related elements of C&DWM. In contrast, in developed countries, C&DWM is more closely linked to stakeholders and shareholders, with slow technological progress and inadequate training being key barriers [18,81].

Technical Means of Recycling Building Materials

- The foundation of the research

Compared to the early innovations in biomaterials, the analyses cited indicate that materials such as waste rubber, PET as a component of fRCA, and GeoPC have demonstrated greater reliability and feasibility [166]. The stockpiling and incineration of waste rubber tires is a major source of carbon emissions, with approximately 1.5 billion tires produced annually worldwide—a number set to increase as population and vehicle numbers grow [4,167]. Shredding waste tires and incorporating them into fRCA has become a primary method for repurposing them as recycled materials in construction, with applications spanning from the building's structure to its infrastructure [168]. Additionally, this process offers good chemical resistance, making it an ideal material for mitigating the effects of hydrochloric acid, sodium sulfate, and chloride ions in marine-climate buildings [169]. However, the presence of heavy metals in tires presents a risk of environmental contamination through concrete leachate, posing a significant challenge in this field [170]. In addition to improvement methods such as HS and GC, incorporating PET fibers is also an effective way to enhance the mechanical properties of RCA. The inclusion of PET in RCA has shown minimal chemical contamination, with the addition of 0.25–10% PET fibers improving concrete's compressive strength by 10–20% and flexural strength by 5–15%. PET is mainly sourced from materials such as copper foils, plastic bottles, and tire cords [171,172]. However, when more than 10% PET is used as coarse aggregate, it diminishes concrete's flexural and ductile properties. Some studies suggest that this percentage should be limited to between 0.25 and 2% for optimal results [173].

- Research response and progress

The literature analysis regarding technological means shows that GeoPC, the most promising alternative to Ordinary Portland Cement (OPC) in the future, has been more widely and maturely utilized in the past decade [103]. GeoPC refers to the dissolution of silica–alumina-based raw materials in alkaline solutions, releasing silica–alumina ions, which undergo a polymerization reaction that enhances the structural strength and sustainability of the concrete unit structure [174]. GeoPC is crucial for reducing the carbon footprint of buildings, with silica–alumina raw materials including red mud (RM), silica fume (SF), rice husk ash (RHA), and FA, among others. The chemical reaction requires alkaline solutions, such as sodium silicate (Na_2SiO_3), potassium silicate (K_2SiO_3), and sodium hydroxide (NaOH). GeoPC reduces the carbon footprint of buildings through the dissolution of these silica–alumina raw materials in alkaline solutions. Moreover, GeoPC achieves consolidation under its own weight, offers better corrosion resistance, and provides superior mechanical properties compared to OPC, with CO_2 emissions reduced by a factor of five to six [175]. Notably, recycled steel fiber (RSF) has garnered significant attention, as it improves the bending performance and impact load resistance of concrete. The primary source of RSF is waste tires, and its recycling method involves low-temperature treatment and chopping, with its physical properties comparable to industrial steel fibers [107]. However, RSF is more prone to corrosion in chloride-rich environments, posing a significant challenge for its future applications [176].

5. Discussion

This study outlines four stages in the development of recycled building materials, from the housing crisis triggered by the dramatic increase in global population to the current climate change concerns due to carbon emissions. Each breakthrough in recycled materials has had a profound impact on the development of the entire construction industry. Notably, after the signing of the Paris Agreement in 2015, the number of publications on recycled building materials surged, accompanied by a significant increase in the diversity of keywords and areas covered. In terms of value strategies, the current research on the CE, LCA, and C&DWM reveals the involvement of multiple stakeholders and the diversified characteristics of the current assessment and decision-making systems, which are also characterized by competing interests and slow progress in some countries and regions. Technologically, the application of C&D waste, waste tires, and PET in RCA has become relatively mature, while advancements in biochar, AWs, GeoPC, and other innovations highlight the multidisciplinary and cross-field characteristics of future recycled building materials.

5.1. Research Patterns

In terms of the global research landscape for recycled building materials, the extent to which different countries are contributing to the field is closely related to policy, economics, and market demand. Among the top five contributing countries, the U.S. Environmental Protection Agency (EPA) has introduced regulations like the *Resource Conservation and Recovery Act* (RCRA) to standardize and regulate the definition, classification, and reuse of recycled materials [177]. The U.S. also has a strong research base, with significant advances in low-carbon concrete and recycled building materials at institutions such as the Massachusetts Institute of Technology (MIT) [178,179]. Australia, affected by geographic isolation and high import costs, has strong demand for recycled materials. In response, the government has introduced high landfill taxes, such as AUD 150 per ton of C&D waste in Sydney, boosting the competitiveness of recycled materials [180]. The market demand, policy regulations, and leading research institutions in these countries contribute to their dominant position. Italy's Italia Domani program introduced the *Strategia Nazionale per l'Economia Circolare*, aiming for an 85% recycling rate for construction waste by 2035, with extensive CE strategies and incentives for energy efficiency in buildings [181]. Spain proposed a 70% C&D waste recycling rate in the 2016 *Plan Estatal Marco de Gestión de Residuos* (PEMAR) and recommended using at least 5% RCA in private construction projects [182]. China, one of the largest C&D waste producers, has experienced significant growth in research due to its large market demand [183]. However, its lack of mandatory recycling quotas and regional inequalities hinder the implementation of recycled construction materials, explaining its high ranking in contribution but low impact [184]. Global research on recycled building materials reveals significant inequalities in technology, policy, market, and economics, which hinder development. This highlights the importance of global cooperation and technology sharing moving forward.

5.2. Markets and Supply Chain

The market and supply chain environments for recycled materials are equally critical, and most studies advocating technological advances and the CE tend to overlook the significant impact of the business environment. Although the 3R principle of reducing, recycling, and reusing has become a broad consensus in C&D waste management, the implementation of this principle depends heavily on business investment and infrastructure development [185]. The first challenge is the incorporation of new operational costs, as C&D waste recycling requires specialized facilities like crushers and screening equipment,

and maintaining the storage and loading of crushed products is costly [186]. Secondly, many unregistered and informal recycling workshops in most countries not only affect the quality of recycled materials but also make it harder for registered facilities to source high-quality materials [187]. Additionally, most recycling enterprises face economic pressures, making it difficult to invest in technological advancements while maintaining basic operations, creating a vicious cycle in the recycling market [188]. Some studies suggest building a closed-loop industrial chain by establishing continuous production lines at suitable locations to enable self-production and sales, reducing transportation and storage costs [189]. Given the global carbon emissions crisis and the financial challenges faced by many developing countries, directly integrating non-registered recycling enterprises and individuals into the supply chain appears to be a more feasible option. For instance, in South Africa, waste pickers in some cities have been allowed to collect waste directly from landfills [190], and in Chile, non-registered individuals or organizations collect waste in the form of cooperatives or autonomous work [191]. Despite the limited technical capacity and scale of recycling by informal organizations, their integration into supply chains alleviates fiscal constraints, reduces time and economic costs for governments, and diminishes exploitative labor relations. This also raises important issues in employment, class, human rights, and the environment, warranting further consideration [192].

5.3. Future Research Trends

5.3.1. Advances and Applications of AI/ML

Firstly, the integration of AI and ML is set to bring transformative changes to the value assessment and performance analysis of recycled building materials. As the predominant assessment method for recycled building materials, LCA faces persistent challenges related to data quality. For instance, inconsistencies arising from the use of different functional units or models can lead to unreliable data comparisons, thereby affecting decision-making accuracy [193]. Moreover, the absence of standardized data collection methods for material inventories further exacerbates these issues [194]. Additionally, traditional manual analysis of electron microscope images suffers from inefficiencies, susceptibility to errors, and subjective bias [195]. Given the current technological landscape, AI and ML represent the most promising solutions moving forward. Regarding data sources, building information modeling (BIM) effectively integrates digital representations of building materials, structures, and energy consumption and is widely recognized as a key information processing method in the global construction industry [196,197]. The incorporation of AI and ML enhances BIM's efficiency in data management, construction, maintenance, and intelligent prediction. Furthermore, for older buildings that were not initially modeled using BIM, AI can rapidly process laser scanning data, classify structural components and surface materials, and facilitate BIM reconstruction. This capability is critical for evaluating building renewal feasibility and assessing the regeneration potential of demolition waste [198]. In terms of workflow, AI can predict the environmental impact of a building during its design phase and conduct LCA to optimize construction strategies while also generating effective BMPs for future reference.

Specifically, the current mainstream AI algorithms applied to LCA include Artificial Neural Networks (ANNs), Convolutional Neural Networks (CNNs), and Long Short-Term Memory Networks (LSTM) [199]. Among them, an ANN has a wide range of applications, including information classification, regression analysis, and time series prediction, and is particularly effective at identifying connections within complex LCA data [200,201]. A CNN is effective at analyzing images for building material identification and layout analysis, making it well suited for early-stage BIM of buildings [202]. An LSTM, a type of recurrent neural network, is suitable for processing time series data and is

often used to predict building energy consumption dynamics and future environmental impacts. For example, in 2024, Płoszaj-Mazurek and Ryńska [203] developed a CNN-based tool to estimate a building's carbon footprint by incorporating factors like local climate and development costs from the user's base model. The core advantage is its ability to estimate carbon emissions early in the architectural and urban planning process. Recent advancements in AI and ML have led to the concept of Construction 4.0, which integrates industrial IoT, cloud computing, big data analytics, and AI technologies. This system offers significant potential for building material sustainability, energy prediction, and full-cycle management, highlighting the value and promising application prospects of AI/ML technologies [204,205].

However, in AI-integrated LCA, current AI and ML models are still limited to providing relatively coarse and imprecise predictions, necessitating manual verification and optimization [206]. Future research should focus on enhancing computational power and developing more reliable data processing models to improve accuracy in this field. Notably, beyond decision making and evaluation, AI can expedite the analysis of X-ray diffraction (XRD) and scanning electron microscope (SEM) images of recycled building materials. This enables rapid identification of mineral compositions and crystalline structures, as well as quantification of porosity and particle size distribution, ultimately aiding in the prediction of material properties in construction applications [207–209]. For example, Pande, et al. [210] developed a multi-objective genetic algorithm with a dynamic fitness function that balances environmental impacts and mechanical properties in real time during material development, significantly enhancing R&D efficiency. Notably, ML agent models capable of rapidly predicting material properties have enabled reverse material design using generative AI [26,211]. Specifically, researchers can input desired material properties and generate new structures or compounds to achieve them. This approach facilitates the exploration of complex physical, chemical, and natural phenomena, accelerating the development of innovative material solutions [212–214]. Additionally, in the context of C&D management, AI-powered image recognition technology can swiftly classify and sort C&D waste by type and size, facilitating efficient material recycling processes [25,215]. In conclusion, AI significantly enhances the efficiency of decision making and research on recycled building materials and represents the most promising innovation direction in this field.

5.3.2. Habitat Improvement

Secondly, future research on recycled building materials should prioritize improving the quality of human habitats. Many conventional construction materials negatively impact indoor and outdoor environments to varying degrees, even when their use remains legally permissible [216]. For instance, polyvinyl chloride (PVC), commonly used in doors, windows, and flooring, releases highly toxic and bioaccumulative dioxins and furans during processing [217]. Similarly, wood preservatives often contain arsenic and chromium compounds [218,219], while asbestos-based insulation materials are linked to malignant tumors and various cancers [220]. Addressing these health hazards associated with traditional building materials is crucial for the advancement of recycled materials. Beyond reducing emissions and resource waste, ensuring a healthier living environment represents a key criterion for future building innovations. In terms of indoor air quality, incorporating recycled wood, bamboo, and other organic materials into construction can reduce the release of volatile organic compounds (VOCs) by up to 88%, while also effectively adsorbing formaldehyde, benzene, and other harmful chemicals [221]. Additionally, linseed oil serves as a viable substitute for chemical adhesives and wood preservatives, although its long drying time remains a challenge [222]. The adoption of these non-fossil-based recycled

materials significantly lowers indoor air pollution levels, thereby mitigating respiratory diseases and allergic reactions. For insulation, recycled materials such as rubber, PET, animal hair, AWs, and organic residues offer excellent thermal and acoustic performance. The thermal conductivity of insulation panels made from recycled rubber is $0.100 \text{ W}/(\text{m}\cdot\text{K})$, while recycled concrete containing PET achieves $0.034 \text{ W}/(\text{m}\cdot\text{K})$ [223]. These materials effectively reduce heat exchange between indoor and outdoor environments, maintaining stable indoor temperatures—particularly beneficial in cold climates or spaces requiring specialized insulation. Likewise, acoustic panels made from coir fibers, wool, and chicken feathers exhibit outstanding sound absorption coefficients of 0.72–0.80, effectively minimizing outdoor noise and creating a quieter indoor environment [224]. Furthermore, green roofs composed of C&D waste, waste polypropylene, inert loam, and compost have an estimated lifespan of 40–50 years [225]. These roofs provide a sustainable solution to the urban heat island effect by regulating indoor temperatures and capturing airborne pollutants such as NO_2 , SO_2 , O_3 , and PM_{10} , with total deposition reaching up to $85 \text{ kg}/(\text{ha}\cdot\text{yr})$ [226,227]. The integration of these recycled materials not only enhances living comfort but also reduces building energy consumption, marking a critical step toward improving residential environments and advancing the sustainability of green buildings.

5.3.3. Developing Countries and Poor Regions

Furthermore, waste recycling and data management of recycled building materials in developing countries are critical issues that must be prioritized by the construction industry. Developing countries, particularly in Africa, South America, and Southeast Asia, face severe challenges in waste recycling [228]. For instance, South Africa generates over 100 million tons of waste annually, with 90% directly disposed of in landfills, including more than 20 million tons of C&D waste [229]. Similarly, Nigeria and India lack systematic C&DWM policies, leading to the disposal of vast amounts of recyclable waste in landfills, contributing to resource wastage and severe air pollution [230]. The environmental and policy challenges in these regions are often more complex, with conflict-affected areas generating significant amounts of “emergency construction waste” due to wars and natural disasters. In most cases, such waste is managed through uncontrolled dumping or burial, exacerbating environmental degradation [231]. Beyond technological limitations, poor management strategies and low recycling rates further contribute to pollution and declining living conditions. For these regions, rather than emphasizing frequent material innovations and complex technological advancements, establishing effective C&DWM frameworks, material recycling standards, and well-structured administrative systems offers a more viable solution. Reforming economic incentives, tax policies, and industry regulations is essential to formalizing the sector and improving recycling efficiency [232,233]. Additionally, technical assistance and knowledge transfer from developed countries play a crucial role in advancing sustainable waste management in these areas [234].

5.3.4. Interdisciplinary Integration

Finally, the future of recycled building materials will increasingly trend toward interdisciplinary integration. At the strategic level, as previously discussed, disciplines such as ecology, economics, and management contribute to systematic assessment models and strategies for recycled building materials, including LCA, the CE, and strategic industry management. These approaches form a highly integrated disciplinary framework, exemplified by building ecology [235,236]. At the technical level, interdisciplinary collaboration has led to the development of new processes for recycled building materials and has enhanced the ability to predict the structural and physical properties of novel materials. For instance, incorporating PET modified by γ -ray irradiation into concrete as RCA has demonstrated su-

perior chemical–mechanical properties compared to directly added PET. This advancement is the result of close collaboration among materials science, chemistry, civil engineering, and nuclear science [237]. Similarly, novel materials such as GeoPC activated by alkaline solutions and biopolymer matrices derived from flax, starch, and lactic acid highlight the significant role of biological sciences in advancing recycled building materials [238,239]. Recent advancements in additive manufacturing for construction have also been driven by the integration of computing, robotics, and automation technologies. These innovations hold the potential to revolutionize the industry by enabling the direct fabrication of building components from digital models, thereby streamlining material production, reducing labor costs, and minimizing carbon emissions associated with construction [240,241].

5.4. Research Limitations

There are still some limitations in this study. Firstly, we excluded databases such as Scopus and non-English journals from data collection, which may limit the global representativeness of the study, particularly for developing countries and impoverished regions. Addressing this limitation will be a key focus of future research. Secondly, the two software tools used in this study are limited to fixed analysis modes, requiring manual screening for thematic evolution and non-key literature clusters. Future research should focus on developing a more comprehensive and efficient literature analysis process with optimized tool integration.

6. Conclusions

This study employs bibliometric analysis to summarize the development of recycled building materials from 1995 to 2025. The findings provide valuable insights for shaping future research directions and policy formulation in this field, while also facilitating the continued advancement and innovation of recycled building materials. Specifically, this study highlights the following key aspects:

- The development of recycled building materials can be categorized into four distinct stages. The significant increase in publications during the last two stages indicates growing global attention to this field, the establishment of stable and representative research institutions and scholars, and a sustained high rate of literature growth in the coming years.
- Core research themes in recycled building materials encompass LCA and sustainability, biological and natural materials, recycled concrete, recycled asphalt and building infrastructure, C&D waste, and environmental impacts and compounding factors, with biological and natural materials emerging as the foremost area of innovation.
- Current research on recycled building materials is structured around two key dimensions: value strategies and technological advancements. In terms of value strategies, LCA, the CE, and C&DWM have garnered significant attention. In terms of technological advancements, recycled concrete remains the most widely utilized material, with substantial innovations focusing on optimizing the RCA production process and its integration into concrete.
- LCA and the CE have long faced challenges related to data reliability and inconsistent analytical methods, while manual analysis of material micro-images suffers from inefficiencies, high error rates, and subjectivity bias. Moving forward, the integration of AI and ML will drive transformative improvements in efficiency, analytical methods, and assessment models, positioning it as the most critical innovation direction in the field.
- The cumulative toxicity and health risks associated with traditional building materials remain a pressing concern. Beyond stricter standardization and production regulation,

greater emphasis should be placed on the role of recycled materials in improving human living environments, incorporating these aspects into the core evaluation criteria for recycled building materials.

- The challenges faced by developing countries remain severe. In these regions, enhancing government efficiency and industry formalization should take precedence over complex material innovation and production processes. Additionally, interdisciplinary collaboration and knowledge sharing will be key drivers for the future development of recycled building materials.

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Abbreviations

The following abbreviations are used in this manuscript:

Artificial intelligence	AI
Machine learning	ML
Construction and demolition	C&D
Construction and demolition management strategies	C&DWM
Polyethylene terephthalate	PET
Polyvinyl chloride	PVC
Fly ash	FA
Agricultural wastes	AWs
Life cycle assessment	LCA
Attributional LCA	ALCA
Consequential LCA	CLCA
Circular economy	CE
Circularity rebound	CR
Construction materials passport	BMP
Global Warming Potential	GWP
Recycled concrete aggregate	RCA
Fine recycled concrete aggregate	fRCA
Natural aggregate	NA
Hot mix asphalt	HMA
Rubberized asphalt	RA
Waste cooking oil	WCO
Waste engine oil	WEO
Geopolymer concrete	GeoPC
Building information modeling	BIM

Appendix A. Other Publications in Figure 9

Arulrajah, et al. [242] 2013

Farina, et al. [243] 2017

Mohajerani, Burnett, Smith, Markovski, Rodwell, Rahman, Kurmus, Mirzababaei, Arulrajah and Horpibulsuk [4] 2020

Silva, et al. [244] 2019
 Tam, Soomro and Evangelista [91] 2018
 Gálvez-Martos, Styles, Schoenberger and Zeschmar-Lahl [18] 2018

Appendix B. Other Publications in Figure 10

Agrela, Barbudo, Ramírez, Ayuso, Carvajal and Jiménez [147] 2012
 Hoyos, et al. [245] 2011
 Disfani, et al. [246] 2012
 Puppala, et al. [247] 2011
 Arulrajah, et al. [248] 2012
 Disfani, et al. [249] 2011
 Arulrajah, et al. [250] 2011
 Arulrajah, Ali, Piratheepan and Bo [248] 2012
 Arulrajah, Piratheepan, Disfani and Bo [242] 2013
 Arulrajah, Disfani, Horpibulsuk, Suksiripattanapong and Prongmanee [109] 2014
 Cardoso, et al. [251] 2016
 Tam, Soomro and Evangelista [91] 2018
 Farina, Zanetti, Santagata and Blengini [243] 2017
 Giani, et al. [252] 2015
 Gálvez-Martos, Styles, Schoenberger and Zeschmar-Lahl [18] 2018
 Ossa, et al. [253] 2016
 Akhtar and Sarmah [230] 2018
 Silva, De Brito and Dhir [244] 2019
 Jin, et al. [254] 2017

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